WASP-10b

R-1210

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_190103` · created 2026-07-31T07:44:16+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458486.6465 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1723 +/- 0.0076
Transit depth (Rp/Rs)^2	2.97 +/- 0.26 [%]
Orbital Inclination (inc)	89.91 +/- 0.6
Airmass coefficient 1 (a1)	1.212 +/- 0.014
Airmass coefficient 2 (a2)	-0.0946 +/- 0.0026
Scatter in the residuals of the lightcurve fit is	1.18 %
Best Comparison Star	None
Optimal Aperture	4.83
Optimal Annulus	8.74
Transit Duration (day)	0.09584 +/- 0.00093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1723 ± 0.0076 vs 0.1592 (deviation 1.72σ)
Duration fit vs published	0.09584 d vs 0.0946 d (deviation 1.35σ)
Mid-time O–C	-4.9 min
Depth SNR	22.7
Residual scatter	1.18 %

Light curve

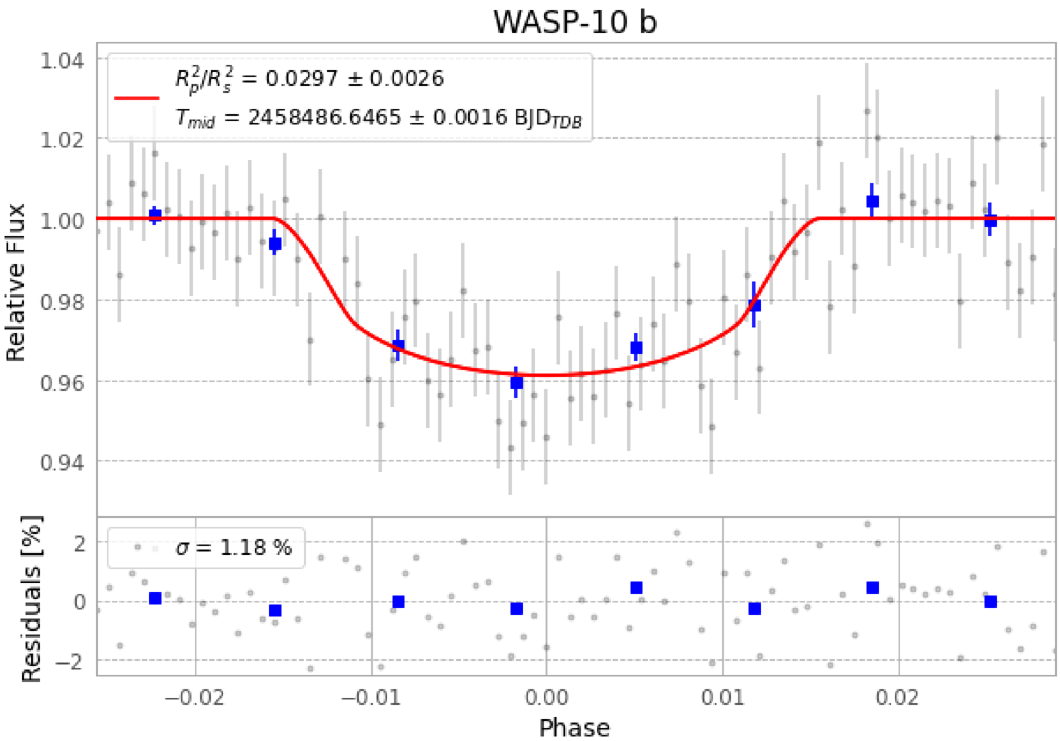

Figure 1 — FinalLightCurve_WASP-10 b_2019-01-02.png (SHA-256 9294eb4dac0337a9…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [286, 341], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e06c2e530a6e4dd6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

82 FITS frames (54 MB), `WASP-10190103013612.FITS` → `WASP-10190103053912.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-190103.log (0e9afd4df289…), FinalLightCurve_WASP-10 b_2019-01-02.png (9294eb4dac03…), FinalLightCurve_WASP-10 b_2019-01-02.csv (d457b095c35d…)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-31 07:44Z.